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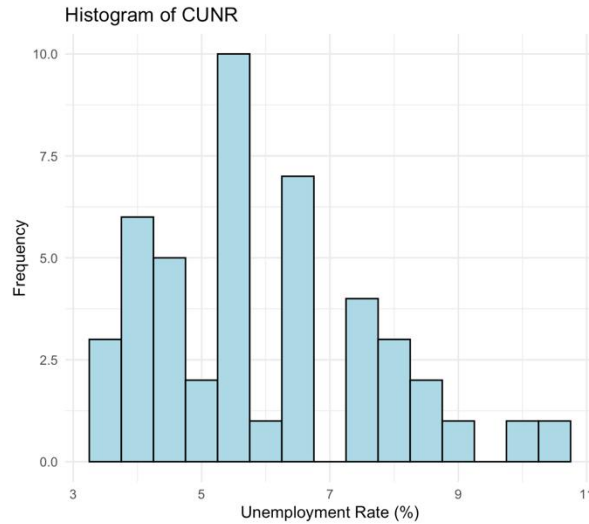
Professor Sarah Abramowitz

Regression and Multivariate Data Analysis, STAT-UB 17

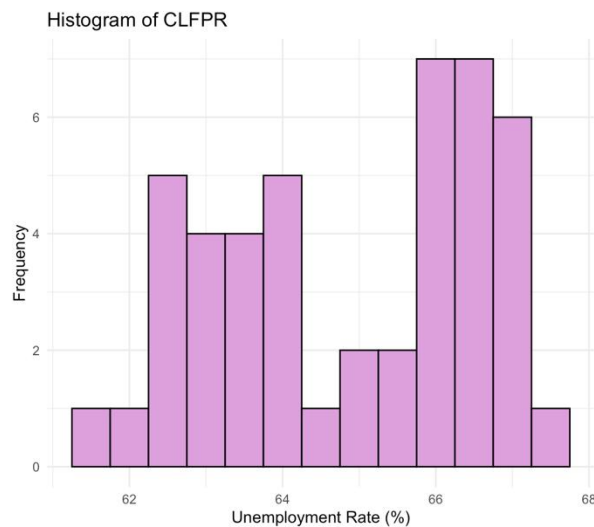
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This report examines whether economic conditions, measured by the civilian unemployment rate, systematically influence labor force participation. In the report, the civilian unemployment rate is denoted as CUNR, and the labor force participation rate is denoted as CLFPR. By analyzing the data from <https://fred.stlouisfed.org/>, we want to determine which of the two economic theories is better supported by the evidence, the discouraged worker effect or the added worker effect. This report begins with a univariate analysis to observe the distribution of both variables. Next, a bivariate analysis of the full range evaluates the overall relationship between the two variables and assesses whether a regression analysis is appropriate. Then, we separate the data into two periods, 1980–2008 and 2009–2024, and apply the bivariate analysis to each period with scatterplots and correlation analysis. These analyses are followed by separate regression analyses and residual analyses to conclude which theory is more consistent with the data.

The unemployment rate has a mean of 5.95% and a median of 5.70%. The mean is slightly bigger than the median, indicating that the distribution is approximately right-skewed but overall symmetric. The 1st quartile is 4.63% and 3rd quartile is 7.15%, indicating that 50% of the observations fall within this range. The standard deviation is 1.71. The histogram below confirms this general shape, showing a slightly longer right tail. Moreover, no extreme outliers are observed in the images.



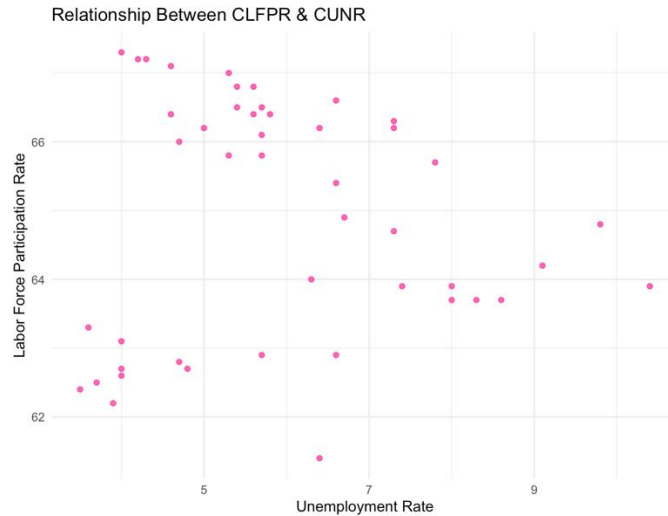
Turning to the labor force participation rate, the participation rate has a mean of 64.89% and a median of 65.15%. Because the mean is slightly smaller than the median, the distribution is a little left-skewed but overall symmetric too. The 1st quartile is 63.4% and the 3rd quartile is 66.4%, indicating that 50% of the observations fall within this range. The standard deviation is 1.72. The histogram below supports this conclusion, which also shows no extreme outlier.



Through preliminary analysis of the two variables, we obtain that the unemployment rate exhibits greater variability than the participation rate since the unemployment rate ranges from

3.50% to 10.40%, while the participation rate ranges from 61.4% to 67.3%. Compared to the unemployment rate, the range of participation rate is relatively narrower.

To visualize the relation between the CUNR and CLFPR for the full range, we draw the scatterplot by using R. This plot illustrates the relationship between the two variables with the CLFPR shown on the y-axis and the CUNR shown on the x-axis. The plot is shown as follows:

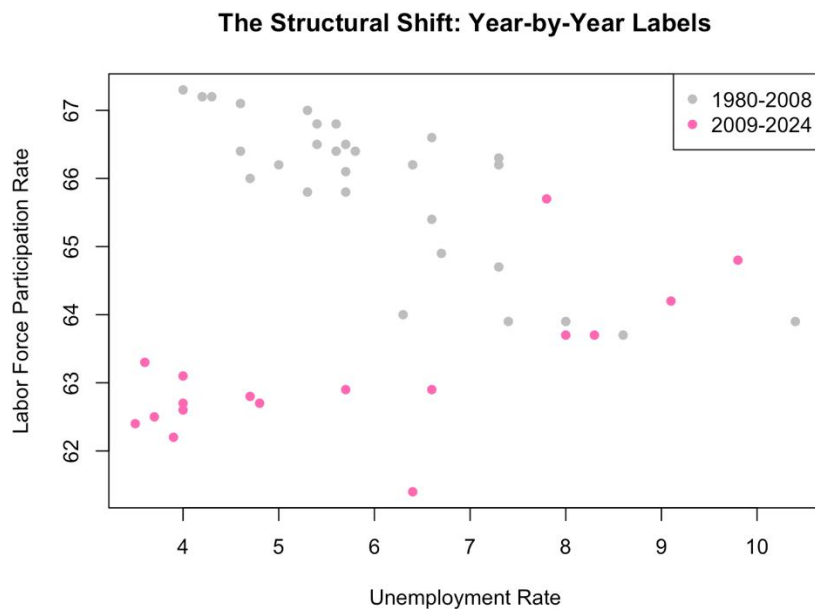


By observing the plot, we cannot find a simple line or curve that gives a good fit for the data. Instead, the observations tend to spread separately without a trend. The absence of this pattern suggests that the relationship is not suitable for linear regression across the entire sample period.

In order to confirm our inference, we apply the correlation analysis for participation rate and unemployment rate, finding that $p - value \approx 0.89 > 0.05$. Therefore, we fail to reject the null hypothesis that the true correlation is zero. Thus, the relationship is not statistically significant, meaning that a single regression model cannot capture the relationship between the two variables.

Since the regression model cannot capture the relationship for the full data range, dividing the whole time into two periods might be a feasible way to find some connections.

Taking 2008 as the dividing line, we divide the time range into two parts: 1980-2008 and 2009-2024. By plotting the two periods on the same scatterplot with different colors, we could observe that a simple curve does not give a better fit for the data than a line does, so a linear model is appropriate for both. In the period 1980-2008, it appears that the CLFPR follows a negative linear relationship. Informally, years with higher unemployment rates tend to have lower participation rates from 1980-2008. After 2008, the situation changed. In the period 2009-2024, it appears that the participation rate follows a positive linear relationship, which is opposite from the previous period. Informally, years with higher unemployment rates tend to have higher participation rates from 2009-2024. Both relationship seems moderate. The scatterplot is as follows:



The correlation test for the period 1980-2008 gives a $p\text{-value} = 3.168e - 07$, which is smaller than 0.05, indicating that the relationship is statistically significant. The value of the correlation between the overall rating and the accountability and transparency score is $r \approx -0.79$. According to Cohen's scale, this value implies that there is a strong negative correlation

between these two variables, suggesting that years with higher unemployment rates from 1980–2008 relatively tend to have lower participation rates. In contrast, for the period 2009–2024, the correlation becomes positive. The $p - value \approx 0.0027$, which is below 0.05, demonstrating that the relationship is statistically significant. The value of the correlation is $r \approx 0.68$, indicating a moderately strong positive relationship. This suggests that in the later period, higher unemployment is associated with higher participation, providing evidence consistent with the added worker effect.

Overall, the correlation signs are opposite, and the values are similar across the two periods, which effectively explains why no statistically significant relationship exists in the data when considering the full range. The opposing relationships in the two eras effectively offset one another, masking the underlying structural change in the relationship between unemployment and labor force participation.

To further investigate, we perform linear regression analysis on the data from both time periods. The regression results for the 1980–2008 period show the p-value of the coefficient on CUNR is $3.17e-07 < 0.05$, indicating that the relationship between CUNR and CLFPR is statistically significant. The estimated regression equation is:

$$CLFPR = 69.66 - 0.62 * CUNR$$

This implies that a 1% increase in the unemployment rate is associated with a 0.62% decrease in the labor force participation rate. The negative slope confirms the negative linear relationship observed in the scatterplot and correlation analysis. The model gives an $R^2 \approx 0.63$, which means that approximately 63% of the variation in CLFPR during this period is explained by unemployment rate alone. According to Cohen, the strength of R^2 is strong. These results provide strong support for the discouraged worker effect during this period.

For the 2009-2024 period, the regression results show the p-value of the coefficient on CUNR is 0.0027, also smaller than 0.05, suggesting that the relationship between the two variables is statistically significant. The estimated regression equation is:

$$CLFPR = 61.28 + 0.33 * CUNR$$

This demonstrates that a 1% increase in the unemployment rate is associated with a 0.33% increase in the labor force participation rate. The positive slope verifies the positive linear relationship observed in the scatterplot and correlation analysis. The model gives an $R^2 \approx 0.46$, which means that approximately 46% of the variation in CLFPR during this period is explained by CUNR. According to Cohen, the strength of R^2 is also strong. These conclusions provide support for the added worker effect, which is contrary to what was obtained in the previous period.

We investigate whether nonlinear transformations would be effective in improving the explanatory power of the model. The scatterplot for the model from 1980 - 2008 indicates that a linear regression analysis is appropriate. There may be a curvilinear component in the model from 2009 - 2024, so log and square root transformations were attempted to improve the model. Both resulted in a lower overall R^2 .

To assess the validity and robustness of the regression model, we apply residual analysis for both periods. For the 1980-2008 period, the standardized residual analysis identifies one outlier: the year 1980 with a standardized residual of $-2.56 < -2$. This demonstrates that the observed labor force participation rate in 1980 was substantially lower than the model predicted. However, when examining influence using Cook's distance, 1983 becomes an influential observation with a Cook's distance of 0.42. This exceeds the threshold value of $4/n$. This

suggests that 1983 is an unduly influential year that has a relatively strong impact on the estimated regression line.

For the 2009-2024 period, the standardized residual analysis recognizes two outliers: the year 2009 with a standardized residual of $2.56 > 2$, and the year 2021 with a standardized residual $-2.6 < -2$. This indicates that the observed CLFPR in 2009 is prominently higher than the model predicted, and the observed CLFPR in 2021 is prominently lower than the predicted value. When examining influence using Cook's distance, only the year 2009 is reported as an influential observation with a Cook's distance of 0.43. This exceeds the threshold value of $4/n$, which suggests that 2009 is an unduly influential year that has a relatively strong impact on the estimated regression line.

The reason why there exist different patterns in the two periods might be the 2008 financial crisis. Before the financial crisis, even if unemployed individuals lost their wage income, they did not necessarily lose all sources of income. Assets such as houses and investments remained part of their financial resources. Consequently, the unemployed often chose to withdraw from the job market. After the financial crisis, however, changes occurred within household structures. Many households experienced significant asset declines. Secondary earners within households, such as homemakers or retired seniors, may have entered the labor market to alleviate financial pressures, creating a positive correlation consistent with the added worker effect. Furthermore, structural shifts like population aging and the gradual rise in female labor force participation may have contributed to this reversal.

The points identified by standardized residuals sometimes differ from those identified by Cook's method, primarily because they measure different aspects. Standardized residuals quantify the deviation between an individual observation's actual value and the regression line's

predicted value, expressed in standard deviation units. Therefore, they can only identify observations with exceptionally large individual prediction errors. In contrast, Cook's distance assesses whether an observation affects the overall slope of the regression model. If an observation lies near the center of the independent variable's distribution, even a large residual will not significantly impact the regression model's slope. Conversely, if an observation's residual is not extreme but its CUNR value is extremely high or low, it will greatly influence the regression line's slope. This is because the regression model must pass near the outliers on both sides. In this dataset, the residual for 1983 is not large, but due to its exceptionally high unemployment rate, it exerts strong leverage, resulting in a large Cook's distance. This explains why the identified point sets do not overlap completely.

To summarize, when examining the full range of data, the information does not provide us with a definitive conclusion. After dividing the data into two periods with 2008 as the cutoff point, analysis reveals that from 1980 to 2008, years with higher unemployment rates tended to have lower labor force participation rates, providing stronger support for the discouraged worker effect. Conversely, from 2009 to 2024, years with higher unemployment rates often exhibited higher labor force participation rates, a finding that better supports the added worker effect.